Taekyung Kim

Research Scientist, NAVER AI Lab

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I study how models learn representations and dynamics of the world and turn them into grounded behavior, from language to physical action, and what internal computations give rise to that behavior.

- **Physical AI:** representation learning, post-training, and analyses for robot control with World Action Models and Vision-Language-Action (VLA) models. [C25, C22, C18, W5, W4]
- **Interpretability of (M)LLMs:** mechanistic analysis of reasoning and information flow across LLMs, VLA models, and omnimodal LLMs. [C24, C20, P1, P2, W3, C11]
- **LLM reasoning and agents:** on-policy distillation, memory agents, and video editing agents (ongoing / under review); mechanistic analysis of chain-of-thought. [C24]
- **Scalable representation learning:** self-supervised representation learning and masked image modeling. [C18, C16, C15, C17, C11]

Professional Experience

Research Scientist, NAVER AI Lab, Korea — Apr. 2022 – Present

Research Intern, Qualcomm Korea — Sep. 2021 – Mar. 2022

Research Intern, Kakao Corp. — Jun. 2019 – Sep. 2019

Education

Integrated M.S. & Ph.D., Electrical Engineering, KAIST — Sep. 2017 – Feb. 2022

- Advisor: Prof. Changick Kim

Integrated M.S. & Ph.D. Program, Mathematical Sciences, KAIST — Mar. 2016 – Aug. 2017

- Advisor: Prof. Mikyung Lim
- Transferred to Electrical Engineering in Sep. 2017

B.S., Mathematical Sciences, KAIST — Mar. 2011 – Feb. 2016

Publications

*: equal contribution, †: co-corresponding author. C: conference, W: workshop, P: preprint.

Conference Publications

[C25] SPARK: Simple Post-training for Adapting pRetrained Knowledge to Robot Control — **CoRL 2026**
Taekyung Kim, Jeongeun Park, Byeongho Heo, Sungjoon Choi, Sangdoo Yun, Dongyoon Han
Also presented at CVPR 2026 Workshop on Scalable Robot Learning Systems

[C24] Beneath the Surface of Chains-of-Thought: A Mechanistic Interpretation of Reasoning Operations in LLMs — **EMNLP 2026**
Seogyeong Jeong, Jaehui Hwang, Dongyoon Han, Geonmo Gu, Alice Oh†, **Taekyung Kim**†

[C23] Isotropic Embedding Perturbations for Robust Vision Language Encoders — **ECCV 2026**
Hyesong Choi, Daeun Kim, Song Park, **Taekyung Kim**, Byeongho Heo, Sangdoo Yun, Dongbo Min, Dongyoon Han

[C22] ORCA: Exploring Conditions for Diffusion Models in Robotic Control — **CVPR 2026**
Heeseong Shin, Byeongho Heo, Dongyoon Han, Seungryong Kim†, **Taekyung Kim**†

[C21] MuCo: Multi-turn Contrastive Learning for Multimodal Embedding Model — **CVPR 2026**
Geonmo Gu, Byeongho Heo, Jaemyung Yu, Jaehui Hwang, **Taekyung Kim**, Sangmin Lee, HeeJae Jun, Yoohoon Kang, Sangdoo Yun, Dongyoon Han

[C20] Map the Flow: Revealing Hidden Pathways of Information in VideoLLMs — **ICLR 2026**
Minji Kim*, **Taekyung Kim***, Bohyung Han

[C19] **Aligned Novel View Image and Geometry Synthesis via Cross-modal Attention Instillation** — **ICLR 2026**
Min-Seop Kwak, Junho Kim, Sangdoo Yun, Dongyoon Han, **Taekyung Kim**, Seungryong Kim, Jin-Hwa Kim

[C18] **Token Bottleneck: One Token to Remember Dynamics** — **NeurIPS 2025**
Taekyung Kim, Dongyoon Han, Byeongho Heo, Jeongeun Park, Sangdoo Yun

[C17] **Masking meets Supervision: A Strong Learning Alliance** — **CVPR 2025**
Byeongho Heo, **Taekyung Kim**, Sangdoo Yun, Dongyoon Han

[C16] **Morphing Tokens Draw Strong Masked Image Models** — **ICLR 2025**
Taekyung Kim†, Byeongho Heo, Dongyoon Han†

[C15] **Learning with Unmasked Tokens Drives Stronger Vision Learners** — **ECCV 2024**
Taekyung Kim†, Sanghyuk Chun, Byeongho Heo, Dongyoon Han†

[C14] **Leveraging Temporal Contextualization for Video Action Recognition** — **ECCV 2024**
Minji Kim, Dongyoon Han, **Taekyung Kim**†, Bohyung Han†

[C13] **HYPE: Hyperbolic Entailment Filtering for Underspecified Images and Texts** — **ECCV 2024**
Wonjae Kim, Sanghyuk Chun, **Taekyung Kim**, Dongyoon Han, Sangdoo Yun

[C12] **Robust Camera Pose Refinement for Multi-Resolution Hash Encoding** — **ICML 2023**
Hwan Heo, **Taekyung Kim**, Jiyoung Lee, Jaewon Lee, Soohyun Kim, Hyunwoo J. Kim, Jin-Hwa Kim

[C11] **What Do Self-Supervised Vision Transformers Learn?** — **ICLR 2023**
Namuk Park, Wonjae Kim, Byeongho Heo, **Taekyung Kim**, Sangdoo Yun

[C10] **Meta Batch-Instance Normalization for Generalizable Person Re-Identification** — **CVPR 2021**
Seokeon Choi, **Taekyung Kim**, Minki Jeong, Hyoungseob Park, Changick Kim

[C9] **Just a Few Points Are All You Need for Multi-View Stereo: A Novel Semi-Supervised Learning Method for Multi-View Stereo** — **ICCV 2021**
Taekyung Kim, Jaehoon Choi, Seokeon Choi, Dongki Jung, Changick Kim

[C8] **Attract, Perturb, and Explore: Learning a Feature Alignment Network for Semi-supervised Domain Adaptation** — **ECCV 2020**
Taekyung Kim, Changick Kim

[C7] **Hi-CMD: Hierarchical Cross-Modality Disentanglement for Visible-Infrared Person Re-Identification** — **CVPR 2020**
Seokeon Choi, Sumin Lee, Youngeun Kim, **Taekyung Kim**, Changick Kim

[C6] **RPM-Net: Robust Pixel-Level Matching Networks for Self-Supervised Video Object Segmentation** — **WACV 2020**
Youngeun Kim, Seokeon Choi, Hankyeol Lee, **Taekyung Kim**, Changick Kim

[C5] **Self-Ensembling with GAN-based Data Augmentation for Domain Adaptation in Semantic Segmentation** — **ICCV 2019**
Jaehoon Choi, **Taekyung Kim**, Changick Kim

[C4] **Self-Training and Adversarial Background Regularization for Unsupervised Domain Adaptive One-Stage Object Detection** — **ICCV 2019**
Seunghyeon Kim, Jaehoon Choi, **Taekyung Kim**, Changick Kim

[C3] **Diversify and Match: A Domain Adaptive Representation Learning Paradigm for Object Detection** — **CVPR 2019**
Taekyung Kim, Minki Jeong, Seunghyeon Kim, Seokeon Choi, Changick Kim

[C2] **Pseudo-Labeling Curriculum for Unsupervised Domain Adaptation** — **BMVC 2019**
Jaehoon Choi, Minki Jeong, **Taekyung Kim**, Changick Kim

[C1] **CNN-Based Semantic Segmentation Using Level Set Loss** — **WACV 2019**
Youngeun Kim, Seunghyeon Kim, **Taekyung Kim**, Changick Kim

Workshop Papers

[W5] **Retrieve, Don't Retrain: Extending Vision Language Action Models to New Tasks at Test Time** — **NeurIPSW 2026**
Jeongeun Park, Juhan Park, **Taekyung Kim**, Sungjoon Choi, Dongyoon Han, Sangdoo Yun
Accepted to NeurIPS 2026 RoboWM Workshop

[W4] **Learning Transferable World-Action Models from Task-Paired Human Videos** — **CoRLW 2026**
Jeongeun Park, **Taekyung Kim**, Dongyoon Han, Sangdoo Yun
Accepted to CoRL 2026 Scaling Human2Robot Workshop

[W3] **VisualScratchpad: Inference-time Visual Concepts Analysis in Vision Language Models** — **ICLRW 2026**
Hyesu Lim, Jinho Choi, **Taekyung Kim**, Byeongho Heo, Jaegul Choo, Dongyoon Han
ICLR 2026 Workshop on Principled Design for Trustworthy AI

[W2] **ActiveNeuS: Active 3D Reconstruction using Neural Implicit Surface Uncertainty** — **AAAIW 2026**
Hyunseo Kim, Hyeonseo Yang, **Taekyung Kim**, YoonSung Kim, Jin-Hwa Kim, Byoung-Tak Zhang
AAAI 2026 Workshop on Artificial Intelligence with Biased or Scarce Data

[W1] **Neural Transformation Network to Generate Diverse Views for Contrastive Learning** — **CVPRW 2023**
Taekyung Kim, Debasmit Das, Seokeon Choi, Minki Jeong, Seunghan Yang, Sungrack Yun, Changick Kim
CVPR 2023 Workshops

Preprints

[P2] **Omni-Modal LLMs Know More Than They Say** — **arXiv 2026**
Na Min An, **Taekyung Kim**, Jaemyung Yu, Hyunjung Shim†, Dongyoon Han†
arXiv:2610.16135

[P1] **Sparse Feature Policy Unlearning Mitigates State Hallucination in Vision-Language-Action Models** — **arXiv 2026**
Jiho Lee, Jeongeun Park, Heayoun Choi, **Taekyung Kim**†, Eunwoo Kim†
arXiv:2610.09496

Patents

Method and Apparatus with Scene Flow Estimation — 2025
Y. Kwak, **Taekyung Kim**, C. Kim, B. Park, C. Park. US Patent No. 12,243,243 B2

Suggesting a New and Easier System Function by Detecting User's Action Sequences — 2024
S. Yun, H. Park, S. Yang, H. Lim, **Taekyung Kim**, J. Choi, K. W. Hwang. US Patent Application US20240045782A1

Mentoring

Research Interns at NAVER AI Lab

Seogyeong Jeong — Mar. 2026 – Sep. 2026
Outcome: [C24] EMNLP 2026

Heeseong Shin — Dec. 2024 – Jun. 2025
Outcome: [C22] CVPR 2026

Minji Kim — Oct. 2023 – Apr. 2024
Outcome: [C14] ECCV 2024

External Mentoring

Jiho Lee — Apr. 2026 – Sep. 2026
Outcome: [P2] arXiv 2026

Research Projects

Development of QA Systems for Video Story Understanding to Pass the Video Turing Test
Institute for Information & Communications Technology Promotion (IITP) Sep. 2017 – Dec. 2021

Monocular Scene Flow Estimation
Samsung Advanced Institute of Technology Aug. 2020 – Jul. 2021

3D Object Recognition Algorithm for Indoor and Outdoor Scenes
LG Electronics May 2019 – Nov. 2019

Professional Service

Conference Reviewer

Computer Vision and Pattern Recognition (CVPR)	2020–2026
Neural Information Processing Systems (NeurIPS)	2020, 2023–2025
International Conference on Machine Learning (ICML)	2021, 2022, 2025, 2026
International Conference on Computer Vision (ICCV)	2023, 2025
European Conference on Computer Vision (ECCV)	2024, 2026
International Conference on Learning Representations (ICLR)	2026
Conference on Robot Learning (CoRL)	2026
ACL Rolling Review (ARR)	2026

Journal Reviewer

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	2020
International Journal of Computer Vision (IJCV)	2020, 2021